

Sharp thresholds and counterexamples for the chemotaxis–logistic trilogy

Beyond-paper results, machine-checked in Lean 4

Formalization: `Shen_work` (Lean 4 / Mathlib v4.29.1)

Abstract

We record the results proved beyond the three Chen–Ruau–Shen chemotaxis–logistic papers. The traveling front’s far-left stability is driven to the sharp threshold $\chi = 4$, with a basin-conditional convergence theorem shown *optimal* by a convective-escape counterexample; and Paper 3’s general- m stability is resolved completely, the sharp divider being a pattern-bifurcation threshold $m_c(U) \approx 2.9$. Every stability statement and the convective-escape counterexample are formalized in Lean 4 (clean-3: only `propext`, `Classical.choice`, `Quot.sound`); the $m > 2$ steady counterexample is established mathematically (Lyapunov–Schmidt, cross-validated numerically) with Lean formalization in progress.

1 Far-left stability of the front: the sharp threshold $\chi < 4$

Normalized co-moving chemotaxis–logistic system ($m = \gamma = \alpha = 1$, $z = x - ct$):

$$u_t = u_{zz} + (c - \chi v_z)u_z + u(1 - \chi v + (\chi - 1)u), \quad -v_{zz} + v = u.$$

Behind the front ($z \rightarrow -\infty$) the solution approaches the plateau $u \equiv 1$; the paper proves stability only for χ near $\chi^* \approx 1$.

1.1 Sharp linear threshold

Linearizing about $(u, v) = (1, 1)$, the dispersion relation ($s = k^2$) is

$$\operatorname{Re} \lambda(s) = -s + (\chi - 1) - \frac{\chi}{1 + s}, \quad \max_{s \geq 0} \operatorname{Re} \lambda = \chi - 2\sqrt{\chi} = -(2\sqrt{\chi} - \chi) < 0 \iff \chi < 4,$$

with critical mode $k_* = 1$ at $\chi = 4$. The general form gives $\chi\gamma < (1 + \sqrt{\alpha})^2$. This coincides with the sharp entropy coefficient $1 - \chi^2/16 > 0 \iff \chi < 4$. (`WholeLineChiPosDispersionSharp.lean: dispersion_le_of_lt_turing`, and the positive/attained companions.)

1.2 A whole-line weighted sharp-entropy engine

With a strictly positive slowly varying weight w , the moving-window relative entropy $E = \int w(u - 1 - \log u)$ satisfies

$$\int w M(u) u_t \leq -\left(1 - \frac{\chi^2}{16}\right) \int w (u - 1)^2 + \frac{C}{\ell} \left(E + \int w(A^2 + W^2)\right) + \text{explicit boundary},$$

$A = u_z/u$, $W = u - 1$, $C = |c| + \chi + 2$. The sharp resolver constant $\frac{1}{4}$ survives the weighting by a real-space integration by parts (no Fourier / multiplier obstruction). (`WholeLineChiPosWeightedEntropyDissipation weighted_sharp_entropy_production.lean`.)

1.3 Basin-conditional convergence, and its optimality

Theorem 1 (basin-conditional, formalized). *For every $0 \leq \chi < 4$, a global front-like orbit that remains in the plateau basin (a uniform far-left lower band) converges to $u \equiv 1$ on the moving far left.*

(...*FarLeftBasinConditional.lean*; *uniformCoMovingLeftEquilibriumConvergence_chiPos_upto_four_ba*.)

Proposition 1 (unconditional $\chi < 4$ is false; formalized counterexample). *Let (U, W) be a traveling wave of lab speed s , and view it in a faster frame $z = x - ct$, $c > s$: $q(t, z) = U(z + (c - s)t)$ solves the co-moving system exactly, with $q(t, z) \rightarrow 1$ as $z \rightarrow -\infty$ (each slice positive at left), yet $q(t, R) = U(R + (c - s)t) \rightarrow 0$ as $t \rightarrow \infty$ for every fixed R . No uniform far-left floor exists. This is convective escape (speed mismatch), already at $\chi = 0$; the closeness hypothesis is exactly what pins c to the wave speed. (*WholeLineChiPosFloorQuantifierObstruction.lean*, *WholeLineConvectiveEscapePDE.lean*.)*

Thus basin-conditional $\chi < 4$ is the optimal statement. The L^1 -mass, bare- L^2 , deep-hole-KPP, front-anchor, and Harnack routes all fail for reasons recorded in the notes.

2 General- m stability (Paper 3, Theorem 2.5): the complete picture

Interval Neumann chemotaxis with u^m mobility, minimal equilibrium on the physical-mass hyperplane. The full m -dependence:

$1 \leq m \leq 2$, in-range χ_0	TRUE (proved)
all $m > 1$, small χ_0	TRUE (proved)
$m \gtrsim m_c(U) \approx 2.9$, large $\chi_0 < \chi_{\text{lin}}$	FALSE (counterexample)
$\chi_0 > \chi_{\text{lin}}$	false (linear instability)

Theorem 2 (proved, Lean). • $1 \leq m \leq 2$: *intervalDomainM_Theorem_2_5_EventualGlobalStabilityFormula* (subcritical + critical $m = 2$, both minimal disjuncts). • all $m > 1$, small χ_0 : *intervalDomainM_Theorem_2_5_sup* via the good-slice bound $\liminf_t \int u^{-2m} u_x^2 \leq \chi_0^2 \mu \sigma_\beta^2$ controlling the oscillation of u^{1-m} .

Proposition 2 (false for $m > m_c$; steady counterexample). *The first Neumann branch's Lyapunov-Schmidt direction coefficient has leading dependence a negative quadratic in m ; for $m \gtrsim m_c(U)$ the bifurcation is backward, producing positive nonconstant mass-preserving steady states for $\chi_0 < \chi_{\text{lin}}$. Starting there never converges to the constant. Explicit: $m = 3$, $\beta = \gamma = \mu = \nu = u_* = 1$, $\chi_{\text{lin}} = 2(1 + \pi^2)$. Cross-validated numerically (pseudo-arclength, residual $\sim 10^{-11}$): $m_c = 2.867, 2.937, 3.006$ at $U = 0.5, 1, 2$.*

The sharp divider is a *pattern-bifurcation* threshold $m_c(U) \approx 2.9$, not the $m = 2$ limit of the mass-seeded energy method.

3 Errata found by formalization

The formalization uncovered several printed statements that are false or overstated for the papers' own data class. Each is *formally refuted* in Lean and replaced by a proved faithful form (full record: *ERRATA.md*, *OUTSTANDING_MAP.md*).

- **Paper 1, Lemma 4.1 / 4.2 / Remark 4.2.** Refuted (`not_Lemma_4_1`, `not_Lemma_4_2`): the stated hypotheses force $m\kappa \leq 1 / D \geq 1$, and the upper barrier is not C^1 across the interface. Corrected: the away-from-interface supersolution and the $\chi = 0$ subsolution slices are proved.
- **Paper 2, Theorem 1.2.** The raw statement is refuted for $a > 0, b = 0$; the correct guard is $a = 0 \vee 0 < b$ (proved).
- **Paper 2, Theorem 1.3.** The printed `FrontierData` is not constructible (`not_IntervalDomainTheorem13Fr`): the honest minimal residual is carried, and case-(iv) is closed for all $m > 0$.
- **Paper 2, Lemma 2.7.** FALSE as stated (`not_Lemma_2_7_intervalDomain`, counterexample $u = t^{-1}$); the faithful continuous-bound form is proved. Lemma 2.6 likewise replaced by its faithful L^p -mass-continuity form.
- **Paper 3, Theorems 2.1–2.5 “all-time” C^1 decay.** A short-time regularity overstatement (false for $u_0 \in C(\bar{\Omega})$, not C^1 : immediate cusp, only $t^{-1/2}$ smoothing). The proved headlines are the faithful *eventual* forms. Theorem 2.1 Part 1 also required an omitted reaction guard (`Theorem_2_1_corrected_intervalDomainM`).

Integrity

All theorems above are Lean 4 / Mathlib formalizations on the `Shen_work` main branch, with axiom audit [`propext`, `Classical.choice`, `Quot.sound`] and no `sorry/axiom/admit/native_decide`. The Lean formalization of the $m = 3$ steady counterexample (amplitude-factored Banach IFT in the repository’s Wiener algebra) is complete: `minimal_equilibrium_global_stability_false_m3` (clean-3), giving the FALSE side of general- m stability entirely in Lean.